

An introduction to measure theory - T.Tao

Statistical Learning Theory Group

Academic Year 2026-2027

Contents

1	Measure Theory	5
1.1	Prologue: The problem of measure	5

Preface

Excercise 0.0.1. *If $(x_\alpha)_{\alpha \in A}$ is a collection of numbers $x_\alpha \in [0, +\infty]$ such that $\sum_{\alpha \in A} x_\alpha < \infty$, show that $x_\alpha \neq 0$ for countable many $\alpha \in A$.*

Intuition:

- Which also means that the number of positive x_α is countable.
- Divide and conquer: We can split original set (a.k.a scope) to smaller subsets and prove the statement locally to each subset before making out the point. 'make sure that the statement is valid on union operation'
- Countable means a set has finite elements, whereas uncountable set has infinite elements.

Proof. Because $\sum_{\alpha \in A} x_\alpha$ is convergent over A, it suffices to show that the sum must be convergent for every subsets of A. (1)

We split A into (can be overlapping cause that doesnt harm anything) subsets. Let n be a natural integer, A_n be subset of A, which includes indexes such that $x_\alpha \geq \frac{1}{n} \quad \forall \alpha \in A_n$, it is obvious that A_n has no zero elements. We first ensure that $\bigcup_{n \in \mathbb{N}} A_n = A$. That is, for every positive real x_α , we can always find large enough $n = \left\lceil \frac{1}{x_\alpha} \right\rceil \in \mathbb{N}$ that $x_\alpha \geq \frac{1}{n}$, thus $\exists n \in \mathbb{N}, x_\alpha \in A_n \quad \forall \alpha \in A, x_\alpha > 0$. As a result, we have **countably** infinite number of subsets. This extension is important because countability is preserved under union of countable number of sets.

We then prove by contradiction over these sets. Assume that there exists a natural number n such that subset A_n has infinite number of elements $||A_n|| = \infty$. Based on definition of A_n , we have $\sum_{\alpha \in A_n} x_\alpha \geq \frac{||A_n||}{n} = \infty$. This contradicts with (1). Therefore, for every natural number n, A_n has finite number of elements.

Finally, we conclude that the number of positive x_α is countable because $\bigcup_{n \in \mathbb{N}} A_n = A$ and each A_n has finite number of elements. Though it possible that we can prove the statement by using supremum over n, this conclusion is more concrete. $\square$

Excercise 0.0.2. Let A, B be sets (maybe infinite), and $(x_{n,m})_{n \in A, m \in B}$, $x_{n,m} \in [0, +\infty]$ be a doubly infinite sequence of extended non-negative reals. Show that

$$\sum_{(n,m) \in A \times B} x_{n,m} = \sum_{n \in A} \sum_{m \in B} x_{n,m} = \sum_{m \in B} \sum_{n \in A} x_{n,m}$$

Intuition:

- For ternary equation, we will prove 2 binary equalities then bridge them. We can use same strategy for Tonelli's theorem for reference.
- Cartesian over finite sets can be written down to collections of collections.
- Tonelli's theorem: nonnegativity + directed finite approximations

Proof. Let $F \subseteq A \times B$ be a finite subset of Cartesian product of A and B. We define $F_A = \{n \in A : \exists m \in B, (n, m) \in F\}$ and $F_B = \{m \in B : \exists n \in A, (n, m) \in F\}$. It suffices to show that F_A and F_B are finite sets, and $F_A \subseteq A, F_B \subseteq B$. F is then can be written as subset of Cartesian of finite sets (big brain move) $F \subseteq F_A \times F_B$. As a result, we have:

$$\sum_{(n,m) \in F} x_{n,m} \leq \sum_{(n,m) \in F_A \times F_B} x_{n,m} = \sum_{n \in F_A} \sum_{m \in F_B} x_{n,m} \leq \sum_{n \in A} \sum_{m \in B} x_{n,m} \quad (a)$$

We need to prove that:

$$\sup_F \sum_{(n,m) \in F} x_{n,m} = \sup_{F_A, F_B} \sum_{n \in F_A} \sum_{m \in F_B} x_{n,m} \quad (1)$$

$$\sup_{F_A, F_B} \sum_{n \in F_A} \sum_{m \in F_B} x_{n,m} = \sum_{n \in A} \sum_{m \in B} x_{n,m} \quad (2)$$

Lemma 1: If N is finite and $a_{n,M} = \sum_{m \in M} x_{n,m} \in [0, +\infty] \quad \forall n \in N$, then $\sum_{n \in N} \sup_M a_{n,M} = \sup_M \sum_{n \in N} a_{n,M}$

Proof of Lemma 1: For every M, $a_{n,M} \leq \sup_M a_{n,M}$ (definition of supremum), therefore $\sup_M \sum_{n \in N} a_{n,M} \leq \sum_{n \in N} \sup_M a_{n,M}$. On the contrary, we need to prove $\sum_{n \in N} \sup_M a_{n,M} \leq \sup_M \sum_{n \in N} a_{n,M}$ to establish equality. This proof has to solve two cases: (1) if $\sum_{n \in N} \sup_M a_{n,M} = \infty$ then there exists at least one element $a_{n_i, M_i} = \infty$ since N is finite, therefore $\sup_M \sum_{n \in N} a_{n,M} = \infty$ which proves the point, (2) if $a_{n,M}$ is real then M is finite by constraint, by definition of supremum, we can always choose a finite subset $M_n \subset M$ such that $\sup_M \sum_{m \in M} a_{n,M} - \epsilon \leq a_{n, M_n} \leq \sup_M \sum_{m \in M} a_{n,M}$, where gathered sum induces:

$$\sum_{n \in N} a_{n,M} \geq \sum_{n \in N} \sup_M a_{n,M} - \epsilon$$

This is true for every $\epsilon > 0$, therefore $\sup_M \sum_{n \in N} a_{n,M} \geq \sum_{n \in N} \sup_M a_{n,M}$.

For (2), this needs to be explicitly proven through induction. For each fixed n, we can see that:

$$\sum_{m \in B} x_{n,m} = \sup_{M' \subseteq B, M' \text{ finite}} \sum_{m \in M'} x_{n,m}$$

Pack this sum as element and with A, we results in final supremum of finite sum:

$$\sum_{n \in A} \sum_{m \in B} x_{n,m} = \sup_{\substack{F_A \subseteq A, F_A \text{ finite} \\ F_B \subseteq B, F_B \text{ finite}}} \sum_{n \in F_A} \sum_{m \in F_B} x_{n,m}$$

For (1), denoting left and right sides of (1) as S_1 and S_2 respectively, (a) has proven that $\sum_{(n,m) \in F} x_{n,m} \leq \sum_{(n,m) \in F_A \times F_B} x_{n,m} \leq S_2$ for every finite set of F , therefore $S_1 \leq S_2$ (1a). On the other hand, $\sum_{(n,m) \in F_A \times F_B} x_{n,m} = \sum_{n \in F_A} \sum_{m \in F_B} x_{n,m}$. For $F_A \subseteq A$ and $F_B \subseteq B$, we would have $F_A \times F_B \subseteq A \times B$ whose supremum is taken by S_1 , which results in $\sum_{n \in F_A} \sum_{m \in F_B} x_{n,m} \leq S_1$ for every finite F_A and F_B . Therefore, $S_2 \leq S_1$ (1b). As a result from (1a) and (1b), equation (1) is proven.

In conclusion, combining (1) and (2), we will have:

$$\sum_{(n,m) \in A \times B} x_{n,m} = \sup_F \sum_{(n,m) \in F} x_{n,m} = \sup_{F_A, F_B} \sum_{n \in F_A} \sum_{m \in F_B} x_{n,m} = \sum_{n \in A} \sum_{m \in B} x_{n,m}$$

The same proof goes for equation: $\sum_{(n,m) \in A \times B} x_{n,m} = \sum_{m \in B} \sum_{n \in A} x_{n,m}$ □

Chapter 1

Measure Theory

1.1 Prologue: The problem of measure

Exercise 1.1.1. *Exercise 1.1.1*